

Semantic Communication Based on Large Language Model for Underwater Image Transmission

Weilong Chen ^{ID}, *Member, IEEE*, Wenxuan Xu, Haoran Chen, Xinran Zhang ^{ID}, Zhijin Qin ^{ID}, *Senior Member, IEEE*, Yanru Zhang ^{ID}, *Senior Member, IEEE*, and Zhu Han ^{ID}, *Fellow, IEEE*

Abstract—Underwater communication is essential for environmental monitoring, marine biology research, and underwater exploration. Traditional underwater communication faces limitations like low bandwidth, high latency, and susceptibility to noise, while semantic communication (SC) offers a promising solution by focusing on the exchange of semantics rather than symbols or bits. However, SC encounters challenges in underwater environments, including semantic information mismatch and difficulties in accurately identifying and transmitting critical information that aligns with the diverse requirements of underwater applications. To address these challenges, we propose a novel SC framework based on Large Language Models (LLMs). Our framework leverages visual LLMs to perform semantic compression and prioritization of underwater image data according to the query from users. By identifying and encoding key semantic elements within the images, the system selectively transmits high-priority information while applying higher compression rates to less critical regions. On the receiver side, an LLM-based recovery mechanism, along with Global Vision ControlNet and Key Region ControlNet networks,

aids in reconstructing the images, thereby enhancing communication efficiency and robustness. Our framework reduces the overall data size to 0.8% of the original. Experimental results demonstrate that our method significantly outperforms existing approaches, ensuring high-quality, semantically accurate image reconstruction.

Index Terms—Semantic communication, large language model, diffusion model.

I. INTRODUCTION

A. Motivation and Background

UNDERWATER communication is a critical technology with widespread applications in various domains, such as environmental monitoring, marine biology research, and underwater exploration [1]. The ability to transmit multimodal data efficiently and reliably in underwater environments is essential for collecting and sharing vital information. For instance, environmental monitoring relies on underwater sensors and vehicles to track changes in marine ecosystems, detect pollution, and study climate change impacts. Marine biologists use underwater communication systems to observe and understand marine life behaviors, migrations, and interactions [2]. Additionally, underwater exploration missions, including resource extraction and archaeological studies, depend heavily on effective communication systems to ensure the safety and success of operations. The integration and transmission of multimodal data, such as images, videos, and sensory data, are crucial to providing comprehensive insights.

Traditional underwater communication methods include acoustic, optical, and radio frequency (RF) communications [3]. Among them, acoustic communication is the most widely used, providing coverage up to tens of kilometers [4]. However, it suffers from low bandwidth, high latency due to the slow speed of sound in water [5], and vulnerability to noise, attenuation, and multipath effects caused by environmental factors [6], [7]. These limitations reduce signal quality and robustness in complex underwater environments.

Currently, some researchers utilize Semantic Communication (SC) based on artificial intelligence to address communication problems under low bandwidth and significant attenuation [8], [9], [10], [11]. SC is a communication paradigm that focuses on transmitting the meaning or intent behind the information, rather than the raw data. This approach aims to optimize bandwidth usage and improve communication efficiency by transmitting only the most relevant and semantically significant data. By introducing a semantic channel to the communication system [12],

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Weilong Chen and Wenxuan Xu are with the School of Computer Science and Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu 611731, China (e-mail: chenweilong1995@std.uestc.edu.cn; wenxuanxu@std.uestc.edu.cn).

Haoran Chen is with Alibaba, Hangzhou 310023, China (e-mail: chr6192@outlook.com).

Xinran Zhang is with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 611756, China (e-mail: xinranzhang@swjtu.edu.cn).

Zhijin Qin is with the Department of Electronic Engineering, Tsinghua University, Beijing 100190, China, also with the Beijing National Research Center for Information Science and Technology, Beijing 100084, China, and also with the State Key Laboratory of Space Network and Communications, Beijing 100084, China (e-mail: qinzhiqin@tsinghua.edu.cn).

Yanru Zhang is with the College of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China, and also with the Shenzhen Institute for Advanced Study, University of Electronic Science and Technology of China, Shenzhen 518110, China (e-mail: yanruzhang@uestc.edu.cn).

Zhu Han is with the Department of Electrical and Computer Engineering, The University of Houston, Houston, TX 77004 USA, and also with the Department of Computer Science and Engineering, Kyung Hee University, Seoul 446-701, South Korea (e-mail: hanzhu22@gmail.com).

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SC extracts and encodes the semantic information of data at the transmitter end, transmits it through the semantic channel, and successfully decodes the semantic information at the receiver end [13]. Traditional communication systems focus solely on the accuracy of bitstream transmission, ignoring the implicit semantic information behind the data. In contrast, SC can extract semantic information, achieving accurate semantic transmission [14] and reducing the amount of transmitted data, thereby optimizing the use of limited bandwidth.

However, traditional SC faces significant challenges in underwater environments, particularly for semantic information tasks. Firstly, the delay spread from multipath propagation and Doppler shifts from relative motion leads to *semantic information mismatch*. Multipath effects and frequency-selective fading distort signals, making it difficult for traditional SC to accurately extract and transmit semantic content, which can lead to misinterpretation of critical data in applications like underwater monitoring. The Doppler effect further exacerbates this by introducing frequency shifts that disrupt the coherence of semantic data, resulting in errors in applications like marine detection, where accurate positioning and identification are crucial for research. Traditional methods, which rely heavily on a singular recovery mechanism without accounting for the dynamic changes in channel conditions and semantic coherence enhancement, are insufficient to accurately reconstruct meaningful data amidst the complex noise and severe interference of underwater environments.

Secondly, SC faces the challenge of *accurately identifying and transmitting different critical information in alignment with the diverse requirements of underwater applications*. Traditional SC frameworks typically lack the ability to distinguish and prioritize varying types of data within communication channels. Their prioritization mechanisms often rely on pre-defined rules [15], [16], [17], which lack the flexibility to adapt to dynamic and user-specific requirements. As a result, they struggle to manage the complexity of underwater applications, where different semantic content must be accurately identified, prioritized, and transmitted according to the specific requirements of the task at hand. For instance, environmental monitoring may focus on transmitting images of coral reef health, while marine biology might prioritize detecting specific marine species. Non-essential semantic information can interfere with the transmission of important data, potentially leading to misinterpretations or delays that negatively impact underwater operations and decision-making processes.

B. Related Work

1) *Underwater Communication*: Underwater communication technique plays an important role in marine biology observation, oil and gas drilling exploration, and natural disaster monitoring. Traditional underwater wireless communication technologies primarily include acoustic communication, optical communication, and RF communication. Acoustic communication utilizes sound signals for transmission, allowing for the greatest propagation distances, which is currently the dominant technology in underwater communication. In the

unique physical environment underwater, acoustic signals are significantly affected by multipath interference and ambient noise. Equalization techniques are employed to overcome inter symbol interference (ISI) caused by multipath propagation [18], [19]. Acoustic communication is characterized by limited bandwidth and slow data transmission rates. Modems are employed to improve transmission rates, mitigate multipath interference, and enhance noise resistance [20], [21]. To reduce data redundancy, data compression techniques are widely used in acoustic communication, including compressed sampling [22] and compressed sensing technologies [23]. Moreover, acoustic signal propagation is easily affected by underwater physical conditions such as temperature, pressure, salinity, and water density, which could lead to signal attenuation and data distortion. Optical and RF communication can provide higher data rates compared to acoustic communication, but both methods are limited in terms of transmission distance [24]. Optical signals exhibit exponential attenuation in water, moreover, optical communication requires a line-of-sight connection between the transmitter and receiver [25], which can result in high error rates and data loss in water with a high concentration of particulates. RF technology can provide medium data transmission rates, but electromagnetic signals experience significant attenuation in water [26], limiting the long-distance propagation of RF technology. Therefore, existing traditional underwater communication technologies face several significant challenges, including severe attenuation, multipath effects, dispersion, limited bandwidth, low transmission rates, and low robustness.

2) *Semantic Communication*: Thanks to the development of artificial intelligence and communication technologies, researchers have developed SC systems based on deep learning to help address the limited bandwidth and robustness of communication. SC can extract the meaning of data in the semantic domain, filter out irrelevant and unimportant information, and significantly reduce the data volume by further compressing the semantic information while preserving the meaning. Therefore, when bandwidth is limited or the signal-to-noise ratio (SNR) is relatively low, SC systems may still perform well, maintaining robustness and reliability. Xie et al. [9] proposed a Transformer-based SC framework to maximize system capacity and eliminate semantic errors. However, compared to text, image data is semantically richer and more bandwidth-sensitive, making the transmission of image data more challenging. Huang et al. [27] proposed an image semantic encoding method based on Generative Adversarial Networks (GANs). Wu et al. [28] proposed a semantic transmission system for segmenting regions of interest in images. Eirina et al. [29] proposed an image transmission method based on joint source-channel coding. Cao et al. [30] proposed an underwater Metaverse framework utilizing quantum imaging and generative AI to create highly immersive and realistic virtual underwater environments. Zhang et al. [31] also propose a generative AI-aided SC framework that integrates deep generative models for semantic feature extraction and reconstruction. Additionally, to further improve transmission efficiency, task-oriented SC has gained attention by focusing on transmitting task-critical information. Xie et al. [32] proposed a task-oriented multi-user SC framework, extracting semantic

information for tasks like image retrieval, to enable efficient single and multi modal data transmission. Ma et al. [33] developed a task-oriented SC system that only selects task-relevant, interpretable features for efficient and robust transmission. Kang et al. [16] designed a task-oriented SC framework for Uncrewed Aerial Vehicle (UAV) image transmission, enabling energy-efficient transfer of user-relevant images through query-based semantic matching.

While early SC systems focused on traditional deep learning techniques, the recent emergence of large language models (LLM) offers new possibilities for enhancing both the semantic understanding and transmission efficiency of SC systems. Researchers [34], [35] apply LLMs like BART and GPT2 to semantic communication from a purely language-oriented perspective. However, they overlook image data, lacks visual-semantic modeling. Jiang et al. [36] utilizes the LLM to convert all modalities into text for unified representation, which simplifies semantic alignment but overlooks the limitations of textual abstraction for visual data. In addition, their respective reconstruction modules, an encoder-decoder pipeline in [37] and a Vision Transformer-based decoder in [38], struggle to recover fine-grained details under complex or noisy environments, limiting their effectiveness in real-world conditions.

Therefore, it is essential to recognize that traditional SC systems often fail to identify and prioritize important information for different tasks effectively. Furthermore, the complex underwater environment introduces challenges such as semantic information mismatch, which can lead to the loss of critical data. As a result, it is crucial to design systems that not only address these challenges but also ensure that essential information is preserved and accurately reconstructed, even in such demanding conditions.

C. Our Contributions

To address these challenges, we propose a novel SC framework based on Large Language Models (LLMs). Unlike traditional models, LLMs offer significant advantages due to their generalization capability, allowing them to be readily applied across a broad range of tasks within a single model. This versatility makes LLMs uniquely suited for underwater communication, where task requirements may vary and the environment presents multiple sources of noise and distortion. Furthermore, LLMs excel at interpreting user intentions, enabling them to distinguish and prioritize different types of information based on their relevance and importance. This capability allows LLMs to selectively filter and manage critical data according to the specific needs of each task, making them a powerful tool for optimizing data transmission in bandwidth-constrained underwater environments. Our framework leverages visual LLMs [39] to comprehend queries from individuals above the surface and perform semantic compression and prioritization of image data, aligning with the diverse requirements of underwater applications. This method allows for the identification and encoding of key semantic elements within images and selectively transmits high-priority information while applying higher compression rates to less critical regions, enhancing the effectiveness and

relevance of data transmission. On the receiver side, we employ a text LLM recovery mechanism to reconstruct the textual data, and two new ControlNet networks to assist the diffusion model in recovering the image to the original size. This novel LLM-based recovery utilizes the semantic understanding and reasoning capabilities of LLMs to accurately reconstruct the textual data while integrating with visual information to provide collaborative guidance for image recovery. Those recovery mechanisms enhance the semantic coherence to mitigate the information mismatch. This approach not only reduces the overall data size to 0.8% of the original but also significantly improves the resilience of the communication system against noise and signal loss. The main contributions of this paper are summarized as follows:

- We introduce an innovative LLM-based SC framework to enhance underwater communication. This framework can perform semantic compression and prioritization, enhancing data transmission by adapting dynamically to the different requirements of underwater applications.
- The proposed framework integrates two newly designed ControlNet networks and a text-based LLM recovery mechanism with a diffusion model, enhancing the semantic coherence while effectively mitigating information mismatch.
- Our experimental results show that our framework can compress data to just 0.8% of its original size while maintaining high-quality transmission, with the Fréchet Inception Distance consistently staying below 0.68, even under significant noise conditions.

The remainder of this paper is organized as follows. Section II introduces the preliminary including SC and underwater channels. Section III introduces the system model and details the proposed framework, including its architecture and technical components. Section IV discusses the experimental setup and results, highlighting the performance improvements achieved by our method. Finally, Section V concludes the paper and outlines potential directions for future research.

II. PRELIMINARY

A. Semantic Communication

SC focuses on transmitting the meaning of information rather than the raw data itself, significantly improving transmission efficiency, especially in bandwidth-constrained environments such as underwater communication. Unlike traditional communication systems that prioritize accurate bit-level transmission, semantic communication emphasizes task-relevant and meaningful information that aligns with the end-user's goals.

The SC system for image data can be formulated as follows. Given an input image $i \in \mathbb{R}^{C \times H \times W}$ and a natural language query q from individuals to indicate the semantic focus, a semantic encoder $\mathcal{E}$ extracts its key semantic representation s , capturing the critical and compressed information:

$$s = \mathcal{E}(i, q), \quad (1)$$

The s is then transmitted over the communication channel and may suffer from distortion due to noise or bandwidth constraints. At the receiver side, a semantic decoder $\mathcal{D}$ reconstructs the image

as i' based on the corrupted version s' :

$$i_r = \mathcal{D}(s'), \quad (2)$$

where i_r represents the reconstructed image.

B. Underwater Acoustic Communication Channel

A realistic physical communication channel is considered to simulate underwater scenarios better. In a shallow water acoustic propagation environment, to consider the multipath propagation and Doppler spread, we first define the path loss between surface and underwater robot, which can be defined as:

$$A(d, f) = d^\sigma a(f)^d, \quad (3)$$

where σ is the spread factor, d is the distance, f is the signal frequency, and $a(f)$ is the absorption coefficient, as calculated using the Thorp formula [40], measured by dB/km with f in kHz, which is expressed as:

$$10 \log(a(f)) = 0.11 \frac{f^2}{1 + f^2} + 44 \frac{f^2}{4100 + f^2} + 2.75 \times 10^{-4} f^2 + 0.003. \quad (4)$$

Secondly, according to [41], we consider the underwater environmental noise $\mathcal{N}(f)$ which is composed of turbulence, ship, wind, and thermal noises [42], denoted as $\mathcal{N}_1(f), \mathcal{N}_2(f), \mathcal{N}_3(f), \mathcal{N}_4(f)$, respectively. This can be defined as:

$$\mathcal{N}(f) = \mathcal{N}_1(f) + \mathcal{N}_2(f) + \mathcal{N}_3(f) + \mathcal{N}_4(f), \quad (5)$$

and those components can be described as:

$$\begin{cases} 10 \log \mathcal{N}_1(f) = 17 - 30 \log f, \\ 10 \log \mathcal{N}_2(f) = 30 + 20s + \log(f^{26}/(f + 0.03)^{60}), \\ 10 \log \mathcal{N}_3(f) = 50 + 7.5v^{1/2} + 20 \log(f/(f + 0.4)^2), \\ 10 \log \mathcal{N}_4(f) = -15 + 20 \log f, \end{cases} \quad (6)$$

where s is the shipping activity factor between [0,1], and v is the wind speed measured in m/s. Given frequency f , The channel conditions are characterized by the signal-to-noise ratio (SNR), defined as:

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{P_{\text{tx}}(f)}{A(d, f) \cdot \mathcal{N}(f)}, \quad (7)$$

where P_{signal} and P_{noise} represent the average received signal and noise powers, and $P_{\text{tx}}(f)$ is the transmitted signal power spectral density. Due to multipath propagation, the transmitted signal at the receiver undergoes the superposition of the direct (Line-of-Sight, LOS) path and multiple non-Line-of-Sight (NLOS) paths. As a result, the total received power is a combination of these different propagation components. According to [41], the SNR resulting from multipath propagation can be expressed as:

$$\text{SNR} = \frac{P_{\text{tx}}(f)}{\mathcal{N}(f)} \times |H(f)|^2 = \frac{P_{\text{tx}}(f)}{\mathcal{N}(f)} \times \left\| \frac{1}{\sqrt{A(d, f)}} + \sum_{p=1}^{q_1} \frac{\Gamma_s}{\sqrt{A_p(l_{s_p}, f)}} e^{-j2\pi f(1+\Delta f_{s_p})\tau_{s_p}} \right\|^2,$$

$$+ \sum_{p=1}^{q_2} \frac{\Gamma_b}{\sqrt{A_p(l_{b_p}, f)}} e^{-j2\pi f(1+\Delta f_{b_p})\tau_{b_p}} \Big\|^2, \quad (8)$$

where $H(f)$ is the channel transfer function, q is the number of significant non-LOS paths, q_1 is the number of non-LOS paths resulting in surface reflection with reflection coefficient Γ_s . q_2 is the number of non-LOS paths resulting from bottom reflection with reflection coefficient Γ_b , $q_1 + q_2 = q$, l_{s_p} and l_{b_p} are the length of s_p th non-LOS path resulting due to surface reflection and b_p th non-LOS path resulting due to bottom reflection respectively, and $\tau_{s_p} = l_{s_p}/c$ and $\tau_{b_p} = l_{b_p}/c$ is the time delay associated with those two paths, c is the underwater speed of sound (1500 m/s), $e^{-j2\pi f(1+\Delta f_{s_p})\tau_{s_p}}$ and $e^{-j2\pi f\tau_{b_p}}$ capture the effect of the phase delay associated with paths. Here, Δf_{s_p} and Δf_{b_p} denote the Doppler frequency shifts of the s_p -th surface-reflected path and the b_p -th bottom-reflected path, respectively. These shifts arise from the relative motion between the transmitter and receiver.

We use the Rayleigh model to characterize the multipath fading and Doppler shifts in the underwater acoustic channel, where the received signal amplitude follows a Rayleigh distribution due to the absence of a dominant LOS component.

In the semantic communication framework, the semantic representation s obtained from the encoder is transmitted, and its corrupted version s' is received. At the physical layer, s is further modulated into a baseband signal $s(f)$. Passing through the underwater acoustic channel with transfer function $H(f)$, the received signal $s'(f)$ can be explicitly written as

$$s'(f) = H(f) s(f) + n(f), \quad (9)$$

where $n(f)$ is the additive noise with power spectral density $\mathcal{N}(f)$.

Considering the above, there are two key challenges in underwater SC:

- How to address the semantic information mismatch caused by multipath propagation and Doppler shifts, which disrupt the coherence and accuracy of semantic content?
- How to identify and prioritize critical information to meet the diverse requirements of underwater applications and ensure efficient communication?

III. METHODOLOGY

The proposed comprehensive framework is tailored specifically for underwater image transmission, which is shown in Fig. 1. This framework initiates with a query generated by individuals located above the water surface, which is subsequently transmitted to the underwater environment. Upon receiving the query, the underwater transmitter employs a semantic encoder integrated with an LLM-based information prioritization mechanism. This mechanism is designed to identify and prioritize critical parts of the visual information based on the query's context. The essential information is then compressed into a format suitable for transmission through the underwater communication channel. Once transmitted, the information is received and decoded by a semantic decoder with a diffusion model

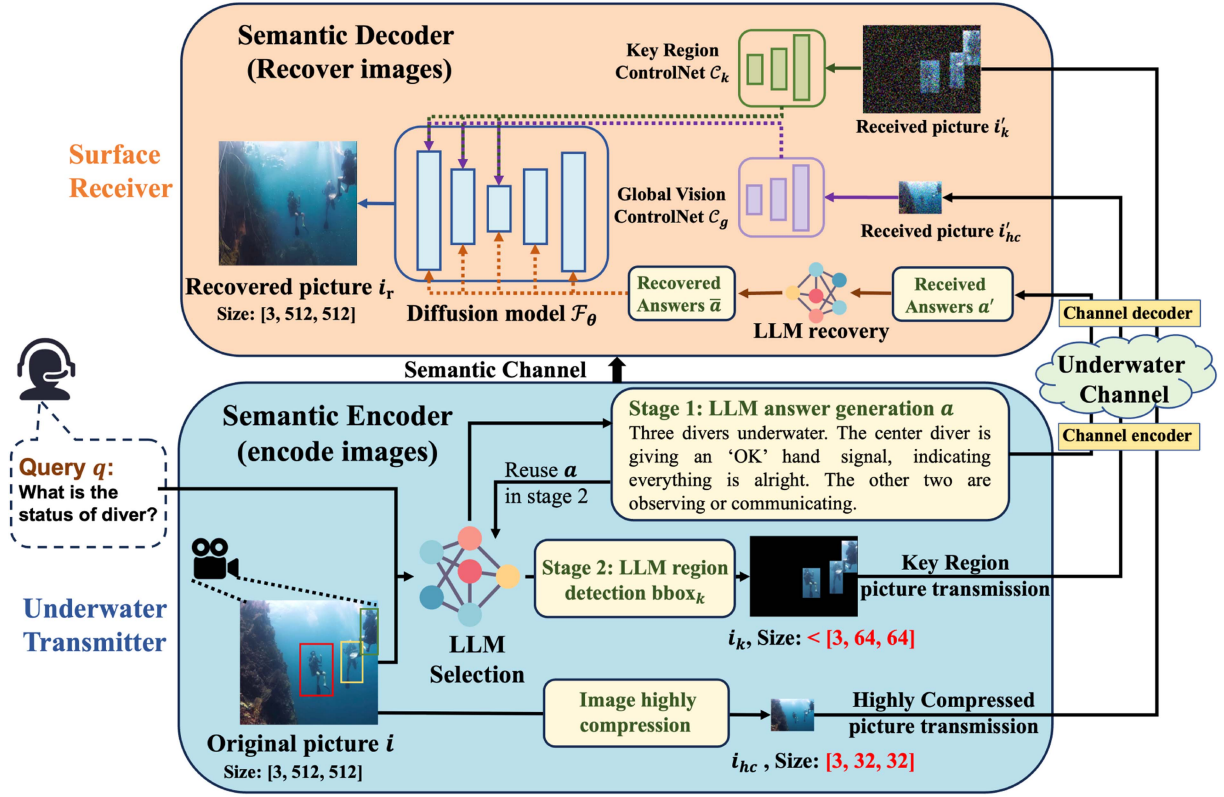


Fig. 1. The framework structure of proposed LLM-based semantic communication.

and LLM recovery at the receiver's end. The decoded visual information is subsequently reconstructed and presented to the individuals above the water. This approach ensures efficient and effective transmission of visual information in underwater communication scenarios.

A. Semantic Encoder

Initially, an individual on the surface provides a query specifying the desired visual information. This query is then transmitted to the semantic encoder by the wireless channel, where we assume lossless transmission as the control signal is lightweight and the focus is on semantic encoding and underwater image transmission. The semantic encoder is composed of two primary components: LLM-based Information Prioritization and Image Highly Compression.

1) *LLM-Based Information Prioritization*: To efficiently transmit images underwater and extract their semantic information, we employ a visual LLM (vLLM) for information prioritization. The process begins with the reception of a natural language query q from individuals above the water surface. These queries serve as the basis for extracting relevant semantic content from an underwater image i . The selection of key regions follows a two-stage process. The first stage is the LLM answer generation stage. Given the query q and the input image i , the vLLM $\mathcal{F}_l$ generates a semantic answer a and identifies the key semantic regions of the image. The query instructs the vLLM

to locate objects or regions of interest in the image, and the corresponding prompt is formulated as:

{Image i } Answer the query according to the image. The query is q .

Here, the Image i placeholder corresponds to the input image and is provided in the vLLM's accepted format, which is typically as a file path string, allowing the model to automatically load and process the image content. The query q is a natural-language sentence designed to ask about specific semantic information within the image, which could be: "What is the status of the diver?" The vLLM processes both the image and the query jointly and outputs a textual response a based on its understanding of the image content. This process can be mathematically described as:

$$a = \mathcal{F}_l(q, i), \quad (10)$$

where a represents the textual response generated by the vLLM based on the query q .

The second stage is the LLM region detection stage. The textual response a is fed back into the vLLM along with the original image i to predict the bounding box bbox_k , which identifies the key region of interest (ROI). The prompt is designed as:

{Image i } Detect the key objective according to the answer in the image. The answer is $\mathbf{a}$.

In this prompt, the Image i is again provided as a file path string, enabling the vLLM to load the same image. The answer $\mathbf{a}$, obtained from the first stage, provides a textual description about the image content, which could be $\mathbf{a} = \text{"The diver is inspecting the pipeline,"}$ then the model is instructed to detect the key objective. The bounding box generation can be mathematically described as:

$$\text{bbox}_k = \mathcal{F}_l(\mathbf{a}, i), \quad (11)$$

where $\text{bbox}_k = [x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ specifies the spatial location of the key region in the image. Once the bounding box bbox_k is determined, the image i_k is cropped from the original image i such that it retains only the pixels within the region defined by bbox_k . The dimensions of the cropped image i_k are determined by the number of channels $C^k = C$, and the height and width, $H^k = y_{\max} - y_{\min}$ and $W^k = x_{\max} - x_{\min}$, derived from the bounding box bbox_k . Specifically, $C^k < C$, $H^k < H$, and $W^k < W$, which helps reduce the image size. Additionally, since only the key regions are transmitted, the data size can be further minimized within the original $\mathbb{R}^{C^k \times H^k \times W^k}$ dimensions, ensuring an even lower transmission overhead. This selective extraction results in a representation that, while retaining the full dimensionality of the original image, significantly reduces informational entropy by excluding extraneous data and preserving only the semantically relevant content. Notably, bbox_k is computed on the original-resolution image i . To further reduce the payload size, the image is resized to a smaller one, and the bounding box coordinates are proportionally scaled to maintain spatial consistency, ensuring that key semantic regions are preserved regardless of their original size.

2) *Image Highly Compression*: To efficiently transmit the entire image with minimal communication bandwidth, we employ image compression techniques to resample and compress the complete image. Resampling involves altering the image resolution, effectively reducing the number of pixels and, consequently, the overall data size. By decreasing the spatial dimensions, we minimize the amount of information that needs to be transmitted, thus reducing bandwidth requirements. Compared to compressing only the non-key regions, this holistic approach offers superior benefits because the image is uniformly compressed. Therefore, it is more efficient to apply compression directly to the entire image rather than separately compressing non-key regions. This strategy ensures that the overall data size is minimized while maintaining an acceptable level of visual quality throughout the image. The highly compressed image i_{hc} can be expressed as:

$$i_{hc} = \text{RESAMPLE}(i), \quad (12)$$

where $i_{hc} \in \mathbb{R}^{C^{hc} \times H^{hc} \times W^{hc}}$. Here, C^{hc} , H^{hc} , and W^{hc} are significantly smaller than C , H , and W , resulting in a substantial reduction in size during transmission. Before transmission, the elements are encoded using a Low-Density Parity-Check

(LDPC) [43] channel encoder. These encoded signals are then transmitted through a Rayleigh fading underwater acoustic channel. At the receiver side, an LDPC decoder is applied to recover the corrupted signals, yielding $\mathbf{a}'$, i'_k , and i'_{hc} , which are then used by the semantic decoder for final image reconstruction.

B. Semantic Decoder

Upon receiving i'_k , i'_{hc} , and $\mathbf{a}'$, the receiver employs a diffusion model $\mathcal{F}_\theta$ to reconstruct the original image. The semantic decoder in the receiver leverages two ControlNets [44], which are specialized neural networks designed to guide the diffusion model in generating controlled images. The diffusion model and two ControlNets are further conditioned by textual information to ensure accurate and detailed image recovery.

The diffusion model $\mathcal{F}_\theta$ belongs to a class of generative models that define a forward and reverse process to generate data. The forward process incrementally adds noise to the data, while the reverse process denoises the data to reconstruct the original input. Formally, the forward process is represented as:

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \quad (13)$$

where $\mathbf{x}_0$ is the original data and $\mathbf{x}_t$ represents the noisy data at time step t . The reverse process aims to recover $\mathbf{x}_0$ from $\mathbf{x}_T$ by progressively denoising the data:

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t), \quad (14)$$

where p_θ represents the learned reverse process parameterized by θ .

To better recover the key regions, we proposed the Key Region ControlNet $\mathcal{C}_k$, which is dedicated to processing i'_k . This network is specifically designed to enhance and accurately generate the critical regions identified during the LLM-based Information Prioritization phase, which can be referred to as:

$$\mathbf{f}_k = \mathcal{C}_k(i'_k), \quad (15)$$

where $\mathbf{f}_k$ is the output of the $\mathcal{C}_k$ and will be integrated into the denoising process in the diffusion model. By focusing on these key areas, the Key Region ControlNet ensures that the significant details, such as semantic segmentation and human pose estimation, are preserved and reconstructed with high fidelity. This targeted enhancement allows for a more refined generation of the most important parts of the image, thereby maintaining the integrity of the prioritized information.

To recover the rest restoration of the image without emphasizing fine details, we proposed Global Vision ControlNet $\mathcal{C}_g$, which is responsible for handling i'_{hc} . The Global Vision ControlNet aims to reconstruct the broader spatial layout and general color distribution of the image, which can be described as:

$$\mathbf{f}_{hc} = \mathcal{C}_g(i'_{hc}), \quad (16)$$

where $\mathbf{f}_{hc}$ is the output of the $\mathcal{C}_g$ and will be also integrated into the denoising process in the diffusion model. The Global Vision

ControlNet $\mathcal{C}_g$ ensures that the reconstructed image retains the correct positioning and overall appearance of the objects within it. This broader approach helps in maintaining the coherence of the entire image while ensuring that the critical regions, processed by the Key Region ControlNet, are accurately integrated into the final output.

In addition to the visual information, we also employ an LLM-based recovery process for the textual guide $\mathbf{a}'$. We also utilize the $\mathcal{F}_l$ as the base model of the LLM recovery process. To recover the text, we use the prompt:

Help me recover the sentence from the noise sentence. The sentence is $\mathbf{a}'$.

In this prompt, the model is asked to recover a corrupted sentence $\mathbf{a}'$ like “Toe dilfe2 is insiect0ng the pi8ellne,” which guides the vLLM to semantically restore the original text from its noisy version, leveraging its language understanding capabilities to correct errors caused by transmission. This process can be defined as:

$$\bar{\mathbf{a}} = \mathcal{F}_l(\mathbf{a}'), \quad (17)$$

where $\bar{\mathbf{a}}$ is the final text utilized by the diffusion model. This step is crucial for preserving the semantic context and ensuring that the textual information is accurately reconstructed. The LLM-based textual recovery leverages the advanced capabilities of large language models to interpret and restore the original textual content, thereby enhancing the overall quality and relevance of the reconstructed information.

To provide fine-grained control over the generative process of diffusion models, we incorporate three guidance signals to enable the generation of more precise and contextually relevant outputs by conditioning the diffusion model on additional information. These guidance signals can be integrated into the reverse diffusion process as follows:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{c}) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, \mathbf{c}, t), \Sigma_\theta(\mathbf{x}_t, \mathbf{c}, t)) \quad (18)$$

where $\mathbf{c}$ represents the control signal provided by the three guidance components, and μ_θ and Σ_θ are the mean and covariance functions parameterized by θ . $\mathbf{c}$ is defined as:

$$\mathbf{c} = \{\bar{\mathbf{a}}, [\alpha \mathbf{f}_k + (1 - \alpha) \mathbf{f}_{hc}]\}, \quad (19)$$

where the hyperparameter α controls the balance between the contributions of the Key Region ControlNet and the Global Vision ControlNet. The complete process of the diffusion model $\mathcal{F}_\theta$, which integrates the outputs from the two ControlNets and the LLM-based textual recovery to generate the final image, can be defined as:

$$\mathbf{i}_r = \mathcal{F}_\theta(\mathbf{x}_t, \mathbf{c}), \quad (20)$$

where $\mathbf{i}_r \in \mathbb{R}^{C \times H \times W}$ is the recovered image. Here, θ denotes the parameters of the backbone diffusion model, and the Key Region ControlNet $\mathcal{C}_k$ and the Global Vision ControlNet $\mathcal{C}_g$ are auxiliary modules attached to the diffusion model, with their own parameters θ_k and θ_g .

By combining these three guidance signals, the diffusion model effectively synthesizes the visual content, ensuring that the reconstructed image is both semantically accurate and visually coherent. The integration of these components allows the model to leverage the strengths of each guidance signal, resulting in a comprehensive and high-quality reconstruction of the original image. The whole process can be found in Algorithm 1.

C. Training Process

To train the semantic encoder, there are question-answering loss and key information prioritization loss. The question answering loss is designed to ensure that the vLLM's answer $\mathbf{a}$ accurately reflects the text label $\mathbf{y}$. This is achieved by using a cross-entropy loss between the predicted answer and the true label:

$$\mathcal{L}_a = -\frac{1}{N} \sum_{n=1}^N \mathbf{y}^{(n)} \log \mathbf{a}^{(n)}, \quad (21)$$

where N is the size of the training data. This loss ensures that the semantic information provided by the LLM is accurately captured and maintained during the reconstruction process.

The key information prioritization loss is designed to ensure that the extracted key regions match the original key regions as closely as possible. To achieve this, we use the bounding box-based loss to ensure that the spatial alignment and dimensions of the key regions are preserved. The bounding box loss is defined as the sum of the Intersection over Union (IoU) loss for the bounding boxes in the original and reconstructed images:

$$\mathcal{L}_{\text{bbox}} = \frac{1}{N} \sum_{n=1}^N \left(1 - \text{IoU}(\text{bbox}_k^n, \text{bbox}_{\hat{k}}^n)\right), \quad (22)$$

where bbox_k^n and $\text{bbox}_{\hat{k}}^n$ are the bounding boxes of the key regions in the original and extracted images, respectively.

To train the semantic decoder, two reconstruction controlnet losses are utilized. For the Key Region ControlNet, Given a set of conditions including time step t , answer $\bar{\mathbf{a}}$, as well as a task-specific condition $\mathbf{f}_k$, Key Region ControlNet learn a network θ_k to predict the noise added to the noisy image $\mathbf{x}_t$ with:

$$\mathcal{L}_k = \mathbb{E}_{\mathbf{i}, t, \bar{\mathbf{a}}, \mathbf{f}_k, \epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_{\theta_k}(\mathbf{x}_t, t, \bar{\mathbf{a}}, \mathbf{f}_k)\|_2^2], \quad (23)$$

where $\mathcal{L}_k$ is the overall learning objective of the diffusion model, $\epsilon \sim \mathcal{N}(0,1)$ is the Gaussian noise injected during the forward diffusion process, $\epsilon_{\theta_k}(\cdot)$ denotes the predicted noise outputs of the Key Region ControlNet. This learning objective is directly used in fine-tuning diffusion models with Key Region ControlNet $\mathcal{C}_k$. The loss of Global Vision ControlNet $\mathcal{C}_g$ can be also written as:

$$\mathcal{L}_g = \mathbb{E}_{\mathbf{i}, t, \bar{\mathbf{a}}, \mathbf{f}_{hc}, \epsilon \sim \mathcal{N}(0,1)} [\|\epsilon - \epsilon_{\theta_g}(\mathbf{x}_t, t, \bar{\mathbf{a}}, \mathbf{f}_{hc})\|_2^2], \quad (24)$$

where θ_g is the network parameters of $\mathcal{C}_g$, $\epsilon_{\theta_g}(\cdot)$ denotes the predicted noise outputs of the Global Vision ControlNet.

To better illustrate the training procedure, we provide Algorithm 2, which summarizes the two-stage training strategy. In Stage 1, only the parameters of the vLLM $\mathcal{F}_l$ are updated, using

Algorithm 1: Proposed LLM-based Semantic Communication.

Require: A query q , underwater image i , pretrained vLLM model $\mathcal{F}_l$, Key Region ControlNet network $\mathcal{C}_k$ with initialized parameters θ_k , Global Vision ControlNet network $\mathcal{C}_g$ with initialized parameters θ_g , diffusion model $\mathcal{F}_\theta$ with pretrained parameters θ .

Transmitter Operations:

- 1: Compute a , bbox_k , and i_k using $\mathcal{F}_l$ according to (10) and (11).
- 2: Resample and compress i to obtain i_{hc} using (12).
- 3: Transmit a , i_k , and i_{hc} through the physical channels to the receiver according to (3), (4), and (5).

Receiver Operations:

- 1: Receive a' , i'_k , and i'_{hc} .
- 2: Compute f_k using $\mathcal{C}_k$ according to (15).
- 3: Compute f_{hc} using $\mathcal{C}_g$ according to (16).
- 4: Recover the text from a' using $\mathcal{F}_l$ according to Eq. (17) to obtain $\bar{a}$.
- 5: Perform the diffusion process and recover the entire image using the diffusion model $\mathcal{F}_\theta$ with f_k , f_{hc} , and $\bar{a}$.
- 6: Obtain the recovered image i_r .

Algorithm 2: Training Process for LLM-based Semantic Communication.**Stage 1: Train the visual LLM $\mathcal{F}_l$ in semantic encoder**

Require: Training dataset with queries q , images i , and ground-truth answers y and bounding boxes bbox_k .

- 1: **for** each training sample **do**
- 2: Predict $a = \mathcal{F}_l(q, i)$.
- 3: Predict $\text{bbox}_k = \mathcal{F}_l(a, i)$.
- 4: Compute loss $\mathcal{L}_a$ and $\mathcal{L}_{\text{bbox}}$ using (21), (22).
- 5: Only update parameters of $\mathcal{F}_l$.
- 6: **end for**

Stage 2: Train the two ControlNets $\mathcal{C}_k$ and $\mathcal{C}_g$ in semantic decoder

Require: Cropped key region i'_k , compressed image i'_{hc} , and answer $\bar{a}$ generated by $\mathcal{F}_l$ and transmitted through the physical channel.

- 1: **for** each training sample **do**
- 2: Freeze $\mathcal{F}_l$ and $\mathcal{F}_\theta$; use $\mathcal{C}_k$ and $\mathcal{C}_g$ get f_k , f_{hc} , and $\bar{a}$.
- 3: Train $\mathcal{C}_k$ with loss $\mathcal{L}_k$ using (23), only update parameters θ_k .
- 4: Train $\mathcal{C}_g$ with loss $\mathcal{L}_g$ using (24), only update parameters θ_g .
- 5: **end for**

question-answering objective $\mathcal{L}_a$ and bounding box regression objective $\mathcal{L}_{\text{bbox}}$. In Stage 2, the vLLM $\mathcal{F}_l$ is initialized with the pre-trained weights from Stage 1 and kept frozen together with the diffusion model $\mathcal{F}_\theta$, while the two ControlNets $\mathcal{C}_k$ and $\mathcal{C}_g$ are trained using the guidance extracted from the previously learned semantic outputs.

IV. CASE STUDY

Here, we present a comprehensive analysis of the performance of the proposed method on the real-world underwater dataset named SUIM Dataset [45]. Specifically, we aim to perform 3 evaluations: (E1) How does the proposed method outperform other approaches? (E2) How does each part contribute to the performance? (E3) How do critical hyperparameters and components impact the proposed method?

A. Dataset Description and Evaluation Metrics

To evaluate the performance of our proposed method, we utilize the SUIM Dataset [45], which comprises over 1,500 images annotated for eight object categories: fish (vertebrates), reefs (invertebrates), aquatic plants, wrecks/ruins, human divers, robots, and seafloor. This dataset has been widely used in research related to underwater image segmentation and enhancement [46], [47], [48], [49]. Based on these categories, we formulate various questions pertaining to the object categories within the images. For the key region identification, we convert the segmentation annotations to bounding boxes corresponding to the object categories. The whole dataset is divided into 80% for training and 20% for testing.

To rigorously evaluate the performance of our proposed method, we employ several metrics: Fréchet Inception Distance (FID) [50], Structural Similarity Index (SSIM) [50], Contrastive Language-Image Pretraining (CLIP) [51], and Learned Perceptual Image Patch Similarity (LPIPS) [52]. FID measures the similarity between the generated images and real images by comparing the distributions of their feature representations extracted from a pretrained Inception network. FID can be defined as:

$$d_F^2(X, Y) = \|\mu_X - \mu_Y\|_2^2 + \text{Tr} \left[\Sigma_X + \Sigma_Y - 2(\Sigma_X \Sigma_Y)^{\frac{1}{2}} \right], \quad (25)$$

where X and Y are the distributions of real image and generated image, respectively, μ_X and μ_Y are the means, Σ_X and Σ_Y are the covariance of X and Y , a lower FID score indicates higher quality and more realistic images.

SSIM assesses the structural similarity between the original and reconstructed images. It evaluates the perceived quality based on luminance, contrast, and structure. SSIM can be defined as:

$$\text{SSIM}(x, y) = [l(x, y)]^\alpha [c(x, y)]^\beta [s(x, y)]^\gamma, \quad (26)$$

where l is the luminance, c is the contrast and s is the structure comparison between the original image x and reconstructed image y . The exponent α, β, γ are positive constants. SSIM values range from -1 to 1 , with higher values indicating better similarity.

The luminance, contrast, and structure comparison of two images can be expressed respectively as:

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}, \quad (27)$$

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}, \quad (28)$$

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x \sigma_y + C_3}, \quad (29)$$

where μ_x and μ_y are the mean values, σ_x and σ_y are the standard deviations, and σ_{xy} is the cross-covariance for two images x and y .

CLIP measures the alignment between images and textual descriptions. CLIP can be defined as:

$$\text{CLIPScore}(x, t) = \max(\cos(x, t), 0) \quad (30)$$

where x is the visual CLIP embedding for an image, and t is the textual CLIP embedding for a description. By evaluating the cosine similarity between the image and text embeddings, we can assess how well the reconstructed images preserve the semantic content described by the text.

LPIPS evaluates the perceptual similarity between the original and reconstructed images. It uses deep features extracted from neural networks to compare image patches, providing a perceptually motivated metric. LPIPS can be defined as

$$d_L(x, y) = \sum_l \frac{1}{W_l H_l} \sum_{h, w} \|w_l \odot (o_{hw}^l(x) - o_{hw}^l(y))\|_2^2, \quad (31)$$

where $o_{hw}^l(x)$ and $o_{hw}^l(y)$ are the activations of a neural network scaled by vector w_l . Then computes the normalized ℓ_2 norm of the weighted activations. Lower LPIPS values indicate higher perceptual similarity.

B. Experimental Setups

During our experiments, we employed a two-stage training strategy to separately optimize the LLM and ControlNet components. The training data included original images of size $C \times H \times W = 3 \times 512 \times 512$, key region images in the size smaller than $C^k \times H^k \times W^k = 3 \times 64 \times 64$, and highly compressed images of $C^{ci} \times H^{ci} \times W^{ci} = 3 \times 32 \times 32$. The LLM was trained to generate bounding boxes that identify key regions in the image based on input queries. Specifically, we fine-tuned the SPHINX-Tiny [53] model using a cross-entropy loss to ensure accurate text-based semantic understanding, combined with a bounding box alignment loss (Intersection over Union, IoU) to align the predicted bounding boxes with ground truth. This training involved pairs of queries and annotated images, where the LLM was guided to output bounding boxes that encompass critical image regions. The batch size for LLM training was set to 4, and optimization was performed using the AdamW optimizer [54] with a learning rate of 1×10^{-4} and weight decay of 1×10^{-6} .

For ControlNet, we trained the Key Region ControlNet and Global Vision ControlNet separately to handle different aspects of image reconstruction. The Key Region ControlNet was trained to refine and reconstruct the key regions, using a mean squared error (MSE) loss between noisy and clean versions of these regions. The Global Vision ControlNet was trained to recover the overall spatial structure and color distribution of the entire image, employing a similar MSE loss on the full image. Both ControlNets are based on Stable Diffusion v1.5 [55],

initialized with pre-trained weights from [44] and fine-tuned using the AdamW optimizer with the same hyperparameters as the LLM training.

In the underwater acoustic communication channel, σ is set at 1.5, s at 0.5, and v at 2 m/s. We employed an acoustic transmitter with a data transmission speed of 60 KB/s. Depending on the SNR, the frequency f and the distance d are adjusted to maintain the desired ratio with respect to the original signal.

Our experiments are implemented based on the PyTorch [56]. All experiments are conducted on a server running Ubuntu 20.04, equipped with a 64-core 2.60 GHz Xeon(R) CPU, 256 GB of RAM, and 4 NVIDIA 4090 GPUs, each with 24 GB of memory.

C. Inference Complexity and Latency Analysis

To evaluate the real-time feasibility of the proposed method, we conducted a theoretical analysis of inference complexity and latency: underwater operations using the Jetson AGX Orin platform and surface-based operations using the RTX 4090 GPU, with a data transmission speed of 60 kb/s.

In the underwater environment, the query, consisting of approximately 20 Bytes, is first transmitted over the underwater acoustic channel. The transmission delay for the query is approximately 2.67 milliseconds. After receiving the query, the SPHINX model (1.1B parameters), quantized to INT8 precision, is deployed on the vLLM [57] to reduce computational overhead. On the Jetson AGX Orin, which has a peak INT8 performance of 275 TOPS, the theoretical throughput is approximately 49,641 tokens per second. For a typical input that generates 200 tokens, the inference latency is 16.1 milliseconds. Therefore, the total latency for the querying process, including both transmission and inference, is approximately 18.77 milliseconds.

Given the reduced size of the cropped key regions after semantic compression, the transmission delay is significantly lower than that of transmitting full-size images. With the average payload size of 52,234 bytes, the transmission delay is approximately 0.85 seconds.

On the surface, the diffusion-based reconstruction module, which includes the Stable Diffusion v1.5 model (1 billion parameters) and two ControlNet models (each with 389 million parameters), operates on an RTX 4090 GPU. With both models quantized to INT8, the total parameter count is approximately 1.78 billion, requiring 3.56×10^{12} FLOPs per image. Since the diffusion-based reconstruction involves 10 denoising steps, the total computational cost is multiplied by 10. Given the RTX 4090's peak INT8 performance of 1400 TOPS, the theoretical inference latency for each step is 2.5 milliseconds, resulting in a cumulative latency of 25 milliseconds for the full reconstruction process.

Combining the results, the proposed method achieves a cumulative computing latency of 0.89 seconds. These results demonstrate that, despite the computational demands of the diffusion process, the method maintains a favorable balance between complexity and real-time performance.

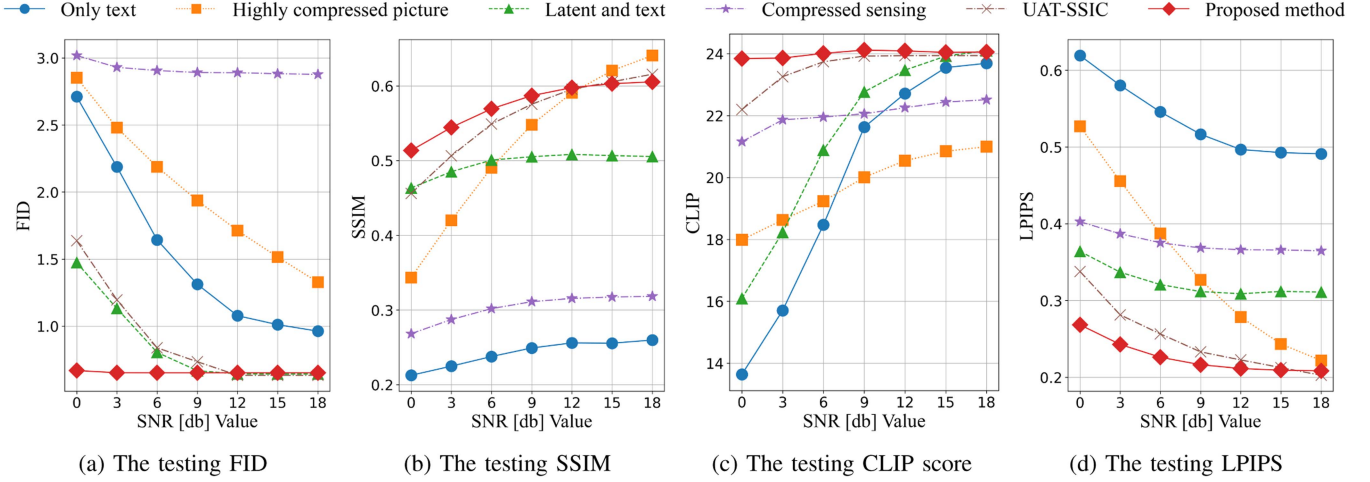


Fig. 2. The testing evaluation of different approaches.

TABLE I

THE PAYLOAD SIZE OF DIFFERENT TRANSMITTED DATA. THE KEY REGION SIZE IS COMPUTED BASED ON THE AVERAGE SPATIAL AREA OF EXTRACTED BOUNDING BOXES ACROSS THE DATASET.

Transmitted data	Dimensionality	Data Type	Payload Size (Bytes $\times 10^3$)
Origin image	[3,512,512]	Float 64	6,291.456
Text	[77]	Int 8	0.077
Highly Compressed picture	[3,32,32]	Float 64	24.576
Key Region picture	$[3,64,64] \times 0.28$	Float 64	27.525
Compressed Sensing	[102,16,16]	Float 64	208.896
UAT-SSIC	[3,64,64]	Float 64	98.303
Proposed method	$[3,32,32] + [3,64,64] \times 0.28 + [77]$	Float 64 + Float 64 + Int 8	52.234 (Reduced 99.2%)

D. Evaluation on Different Approaches (E1)

To simulate real-world underwater scenarios, the experiments are conducted under 0, 3, 6, 9, 12, 15, 18 SNR settings.

1) *Baselines for Comparison:* Our proposed method is compared with the following four approaches:

- *Only text:* Transmit only the text data and recover it directly at the receiver using textual decoding methods.
- *Highly compressed picture:* Transmit highly compressed images and recover them using a picture resampling method.
- *Latent and text:* A novel language-oriented SC framework that transmits both text and a compressed image embedding and is recovered with diffusion model, as described in [58].
- *Compressed Sensing (CS)* [59], [60]: Transmit sparsely sampled data generated using compressed sensing techniques and reconstruct the original image using sparse recovery algorithms.
- *UAT-SSIC* [61]: incorporates a semantic segmentation-based sonar image compression (SSIC) framework along

with a joint source-channel codec to enhance the semantic fidelity of reconstructed sonar images. For consistency with other methods, the original sonar images are replaced by the three-channel standard image format and uniformly compressed with a $64 \times$ compression ratio.

As shown in Table I, the proposed method achieves a significant reduction in payload size, decreasing it by 99.2% compared to the original image. The cropped key region i_k is dynamically determined based on the bounding box $bbox_k$ predicted by the vLLM, and its size varies according to the spatial extent of the detected region. The average key region size accounts for approximately 28% of the original image area. Additionally, the vLLM generates a semantic answer with an average length of 83 characters, contributing minimal overhead to the overall transmission.

To show the performance under different noise scenarios, we evaluate the different metrics under different SNR settings, as shown in Fig. 2. As depicted in Fig. 2(a), the proposed method consistently outperforms the baseline approaches in terms of FID scores, demonstrating its ability to generate high-quality, realistic images. Lower FID scores indicate better visual quality, and our method maintains a significant margin over the others, particularly at lower SNR levels. The high visual quality is due to the effective combination of key region emphasis and holistic image reconstruction. The performance of compressed sensing is poor in terms of FID and does not improve with higher SNR. This is because FID measures the overall visual quality and diversity of generated images, whereas compressed sensing primarily recovers low-frequency components, leading to poor image details and a lack of realistic texture, which results in consistently higher FID scores. UAT-SSIC and the Latent and text method achieve better results than the compressed sensing and highly compressed picture baselines, primarily due to their use of diffusion models for image reconstruction. This enables both methods to produce visually realistic outputs, even under moderate noise conditions.

Fig. 2(b) illustrates that the proposed method attains the highest SSIM scores when SNR is less than 12, signifying

excellent structural similarity with the original images in the highly noisy environment. The high SSIM scores are due to the method's focus on preserving both the key regions and the overall image structure. UAT-SSIC performs slightly lower SSIM scores than the proposed method when SNR is below 12, but achieves comparable scores at higher SNR values, since it can preserve structures well through semantic-aware reconstruction but lacks the explicit all-region enhancement for whole picture recovery. The SSIM for only transmitting the text does not significantly increase with higher SNR values because transmitting only textual information lacks the direct structural cues present in the visual data, making it inherently challenging to reconstruct the original image's structural details accurately. Similarly, the performance of compressed sensing is comparable to that of only transmitting the text, as it too suffers from a lack of detailed structural information. As a result, even as the noise decreases (higher SNR), the structural similarity cannot substantially improve because the text alone and compressed sensing do not provide sufficient information to fully restore the image's spatial and structural attributes. The highly compressed picture method outperforms the proposed method at higher SNR values due to its simpler approach, which benefits more from reduced noise and better preserves certain structural details.

Fig. 2(c) shows that our proposed method achieves the highest CLIP scores under a more noisy environment, indicating superior semantic alignment between the reconstructed images and the textual descriptions. The high CLIP scores for the proposed method are attributed to the effective integration of textual guidance and image reconstruction, ensuring that the reconstructed images closely match the input queries. The Highly compressed picture transmission does not show a steep increase in CLIP scores like the other methods because CLIP primarily measures the alignment between text and images. Since this method does not utilize textual guidance, it cannot leverage the reduction in textual noise to improve semantic alignment as effectively as the other methods. Additionally, the Latent and text transmission surpasses the proposed method at $SNR = 18$ because it transmits more information, with the transmission size being 1.3 times that of the proposed method. UAT-SSIC performs slightly lower than the proposed method in terms of CLIP score since CLIP emphasizes fine-grained semantic-text alignment, which is harder to achieve without explicit textual conditioning.

The LPIPS scores presented in Fig. 2(d) reveal that the proposed method achieves the lowest values across all SNR settings, indicating superior perceptual similarity to the original images. The low LPIPS scores for the proposed method are a result of its ability to preserve fine details and perceptual features during the reconstruction process. UAT-SSIC shows slightly higher LPIPS scores than our method, as it lacks the richer perceptual guidance provided by our two-ControlNet design. The only text transmission exhibits a relatively flat LPIPS curve as SNR increases. This flatness occurs because the reconstruction process relies solely on textual information, which lacks the direct visual features needed to accurately reproduce perceptual details. Similarly, the performance of compressed sensing shows minimal change across varying SNR due to its reliance on low-frequency

components that lack the high-frequency perceptual details needed to improve perceptual similarity. As a result, even when noise is reduced (higher SNR), the lack of detailed visual cues prevents significant improvement in perceptual similarity.

The image illustrating the results of our proposed generation method is presented in Fig. 3, corresponding to the scenario where $SNR = 0$. Despite the heavy noise environment, the proposed method demonstrates superior image reconstruction capability. This effectiveness is due to the innovative use of LLM-based semantic communication principles, where key semantic elements are preserved while unnecessary details are effectively compressed. This approach not only enhances the image's clarity and detail in harsh underwater communication scenarios but also maintains a high level of structural and semantic integrity.

Overall, the results clearly demonstrate the effectiveness of our proposed method in achieving high-quality, semantically accurate, and perceptually similar image reconstructions under various SNR settings, significantly outperforming the baseline approaches.

E. Ablation Study (E2)

In this subsection, we systematically evaluate the impact of different components of our proposed Semantic Communication (SC) framework. Understanding the contribution of each component provides valuable insights into the robustness and adaptability of the framework. For this evaluation, we individually remove the Key Region ControlNet and the LLM recovery mechanism in the semantic decoder to assess their respective contributions. Specifically, removing the Global Vision ControlNet poses a unique challenge as the Key Region ControlNet alone is designed to recover only key regions. Thus, it would be inappropriate to measure the overall performance using standard metrics without the Global Vision ControlNet. To address this, we analyze the hyperparameters in the next section to demonstrate the impact of the Global Vision ControlNet comprehensively.

1) *Evaluation Results:* Fig. 4(a) presents the FID scores for the different configurations. The absence of the Key Region ControlNet and the LLM recovery part results in significantly higher FID scores under high noise conditions, indicating that these components are crucial for maintaining image quality in noisy environments. As the SNR values increase, the FID scores for all methods converge and exhibit less variation because the reduction in noise diminishes the adverse effects of missing components.

Fig. 4(b) shows that the removal of key components leads to a noticeable decrease in SSIM scores, underscoring their importance in preserving structural details in the reconstructed images. Notably, the SSIM scores for the configuration without LLM recovery do not increase as rapidly as those for the proposed method and the configuration without the Key Region ControlNet when the noise decreases. This phenomenon can be attributed to the fact that more accurate text information significantly aids in recovering structural similarity under low noise conditions.

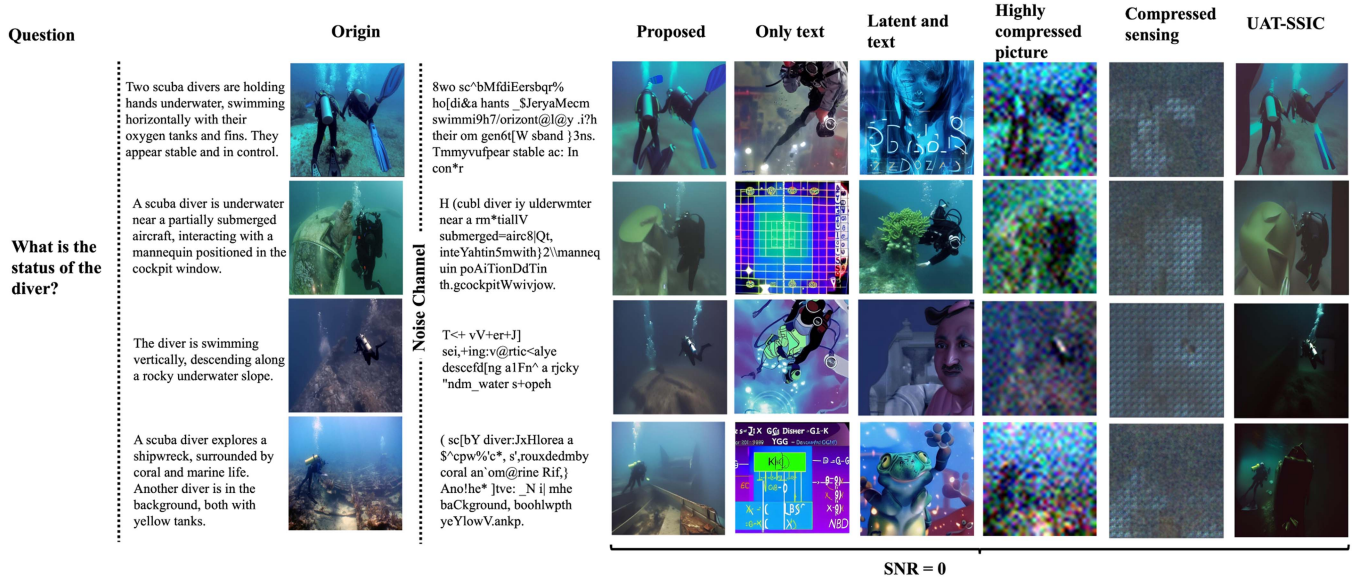


Fig. 3. The generation picture of different approaches when SNR = 0.

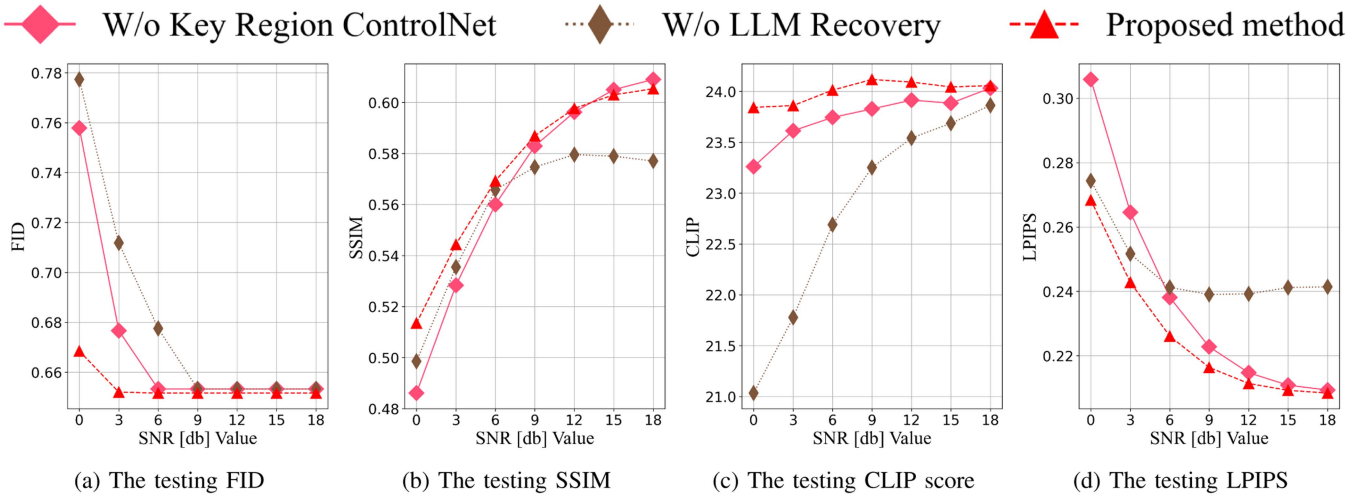


Fig. 4. The testing evaluation of ablation study.

Fig. 4(c) shows that the absence of the LLM recovery part leads to a significant drop in CLIP scores under high noise conditions, underscoring the crucial role of LLM recovery in restoring semantically relevant information. In contrast, removing the Key Region ControlNet results in a less pronounced, yet still noticeable, decline in CLIP scores. This indicates that while the reconstruction of key regions is less critical than LLM recovery for maintaining semantic alignment, it still provides considerable support in preserving language-related coherence.

Fig. 4(d) presents the LPIPS scores for the different configurations. Similar to the SSIM scores, the removal of key components results in higher LPIPS scores under high noise conditions, whereas under low noise conditions, the absence of the LLM recovery has a more pronounced negative impact. In high noise environments, the Key Region ControlNet is

essential for preserving perceptual similarity by focusing on critical image regions. However, as the noise decreases, the significance of the LLM recovery becomes more apparent. The accurate textual information provided by the LLM recovery mechanism significantly enhances perceptual quality, making its absence particularly detrimental in low noise conditions.

F. Evaluation on Key Hyperparameters (E3)

In this subsection, we separately analyze the impact of varying α and Doppler shifts, which control the relative influence of the Global Vision ControlNet compared to the Key Region ControlNet, as well as the robustness of the system to underwater channel dynamics. For simplicity, we use Δf to denote the Doppler shift in general, without distinguishing between Δf_{s_p}

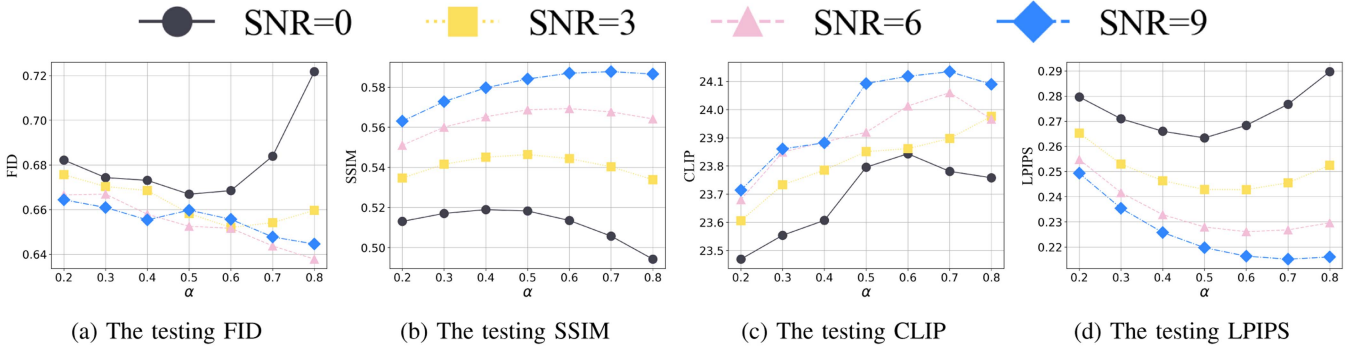
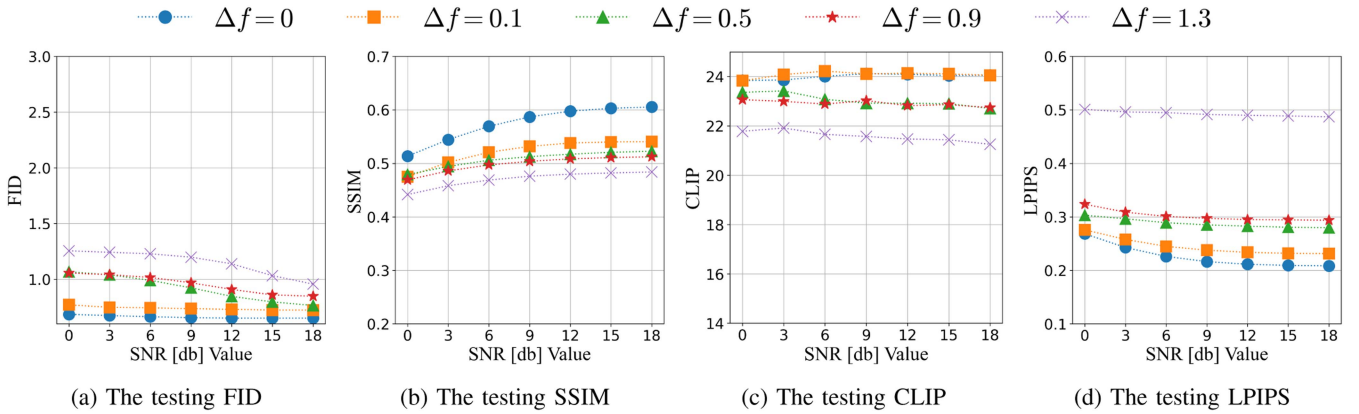
Fig. 5. The testing evaluation of different α .

Fig. 6. The testing evaluation of different Doppler shifts.

and Δf_{b_p} . We evaluate the performance of α values ranging from 0.2 to 0.8 under high noise conditions, specifically for SNR values from 0 to 9. In addition, we examine the effect of different Doppler shifts Δf , ranging from 0 to 1.3, to simulate various levels of transmitter-receiver relative motion. The results are presented in Fig. 5 and Fig. 6.

1) *Evaluation on Different α* : Fig. 5(a) presents the FID scores for different α values. We observe that under extremely high noise conditions (SNR = 0 or 3), the best performance is achieved with α values between 0.5 and 0.6. This is because, in the presence of strong noise, the information capacity of the transmitted f_k (key regions) is greater than that of f_{ci} (contextual information), due to the larger image size of the key regions. Therefore, the impact of the Key Region ControlNet becomes more significant, leading to better results. However, as the noise decreases, the contextual information provided by f_{ci} can work more effectively, reducing the relative benefit of the key regions. Thus, the optimal α value shifts, and the influence of the Key Region ControlNet diminishes. Interestingly, we also observe that FID scores can become higher at lower noise levels. This phenomenon can be attributed to the nature of what FID measures: the quality and diversity of generated images compared to real images. At lower noise levels, the reconstructed images are more detailed, which can reveal more discrepancies between the generated and real images if the reconstruction is not

perfect. Consequently, any small imperfections in the generated images are more noticeable, potentially leading to higher FID scores despite the lower noise. This highlights the sensitivity of FID to subtle differences in image quality and diversity, which can be more pronounced when noise is less of a factor.

Fig. 5(b) and 5(d) show the SSIM and LPIPS scores for different α values, respectively. We observe that the peak SSIM scores for all SNR settings occur around $\alpha = 0.5$ to 0.6 , indicating that both ControlNets contribute optimally to the structural similarity of the reconstructed images at these values. Similarly, the LPIPS scores exhibit a comparable trend, with the best perceptual similarity achieved near $\alpha = 0.5$ to 0.6 . These results suggest that the unique characteristics of SSIM and LPIPS, which measure structural similarity and perceptual similarity, respectively, are best satisfied when both the Global Vision ControlNet and Key Region ControlNet are balanced. The combined influence of these two ControlNets ensures that key regions and overall image context are both effectively reconstructed, leading to higher SSIM and lower LPIPS scores. This balance maximizes the retention of structural details and enhances perceptual quality, demonstrating the necessity of both ControlNets working together to achieve the best possible results.

Fig. 5(c) presents the CLIP scores for different α values. CLIP measures the similarity between images and their textual descriptions, which tend to be more generalized and descriptive.

As α increases, the overall clarity and detail of the image improve, making it more aligned with the textual descriptions. Consequently, the CLIP scores reach their peak around $\alpha = 0.6$ to 0.7, indicating optimal semantic alignment between the reconstructed images and the accompanying text. However, when α is further increased beyond this range, we observe a decline in CLIP scores. This decline suggests that an excessive focus on the Global Vision ControlNet at the expense of the Key Region ControlNet can lead to a loss of specific details that are crucial for maintaining the nuanced alignment between the images and their textual descriptions. In other words, while greater α values initially enhance overall image quality, overly high values may diminish the semantic coherence of key regions, thereby reducing the effectiveness of the CLIP score. This balance underscores the importance of a nuanced approach to optimizing α to achieve the best semantic alignment.

2) *Evaluation on Different Δf* : Fig. 6 demonstrates the reconstruction performance of the proposed system under various Doppler shift conditions $\Delta f \in \{0, 0.1, 0.5, 0.9, 1.3\}$. As the Doppler shift increases, performance on all four metrics FID, SSIM, CLIP, and LPIPS, gradually degrades, reflecting the increasing distortion introduced by the underwater channel. This trend is particularly evident in SSIM and LPIPS, where high Δf values lead to more noticeable degradation in visual and perceptual quality. However, under small Doppler shifts (e.g., $\Delta f = 0.1, 0.5$, and 0.9), the reconstructed images retain high quality across all SNR levels. Despite the presence of significant Doppler effects at larger shifts $\Delta f = 1.3$, the four metrics indicate that the system still preserves the global structure and semantic content of the transmitted images. These results highlight the adaptability of the proposed framework to challenging underwater conditions.

In summary, the evaluation of different α values demonstrates that a balanced contribution from the Global Vision ControlNet and the Key Region ControlNet is essential for optimizing semantic alignment, image quality, perceptual similarity, and structural fidelity under high noise conditions. Furthermore, the analysis across varying Doppler shifts Δf shows that the proposed system maintains strong reconstruction performance when the frequency distortion is moderate. Even though severe Doppler effects degrade performance, the overall semantic structure and visual integrity of the reconstructed images remain largely preserved, confirming the robustness of the system in realistic underwater acoustic environments.

V. CONCLUSION

In this paper, we have introduced a novel Semantic Communication (SC) framework based on Large Language Models (LLMs) to address the challenges of underwater communication. Our framework leverages visual LLMs for semantic compression and prioritization of underwater image data, ensuring the preservation of key semantic elements while significantly reducing the overall data size. By integrating two newly designed ControlNet networks with a diffusion model, our method achieves superior image reconstruction quality, maintaining high perceptual and structural fidelity even under

noisy conditions. Experiments show that our method not only enhances semantic alignment and perceptual quality but also ensures robust and efficient data transmission, reducing the data size to 0.8% of the original. While the SUIM dataset of 1500 images was used to evaluate the framework, we acknowledge the potential benefits of incorporating more diverse underwater datasets in future work to further validate and extend the generalizability of our approach. These findings highlight the potential of our approach to significantly improve the performance of underwater communication systems.

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Weilong Chen (Graduate Student Member, IEEE) received the BS degree, and the MS degree in electronic engineering from the University of Electronic Science and Technology of China, where he is currently working toward the PhD degree with the School of Computer Science and Engineering. He has authored or coauthored papers in top conferences such as ACM MM and journals such as *IEEE Transactions on Knowledge and Data Engineering*, *IEEE Transactions on Smart Grid*, and *IEEE Transactions on Circuits and Systems for Video Technology*. His research interests include machine learning, federated learning, and smart grid.



Wenxuan Xu received the BS degree in computer science and technology from the Southwestern University of Finance and Economics, Chengdu, China. She is currently working toward the MS degree in computer technology with the University of Electronic Science and Technology of China. Her research interests include machine learning, large language models, and smart grids.



Haoran Chen received the bachelor's and MS degrees and from the University of Electronic Science and Technology of China. He is currently a senior algorithm engineer with Alibaba. His recent research interests include large language model (LLM) systems and applications, particularly data synthesis, reward modeling, and reinforcement learning related to agentic capabilities.



Xinran Zhang received the BS degree in information and communication engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2018, the MS degree in electrical and computer engineering from The Ohio State University, Columbus, OH, USA, in 2019, and the PhD degree from the School of Computer Science and Engineering, UESTC. She is currently an assistant professor with Southwest Jiaotong University, Chengdu. Her research interests include federated learning, vehicular network, mobile computing, and

machine learning.



Zhijin Qin (Member, IEEE) is currently an associate professor with the Department of Electronic Engineering, Tsinghua University, Beijing, China. She was with Imperial College London, Lancaster University, and Queen Mary University of London, U.K., from 2016 to 2022. Her research interests include semantic communications and satellite communications. She was the recipient of 2017 IEEE GLOBECOM Best Paper Award, 2018 IEEE Signal Processing Society Young Author Best Paper Award, 2021 IEEE Communications Society Signal Processing for Communications Committee Early Achievement Award, 2022 IEEE Communications Society Fred W. Ellersick Prize, and 2023 IEEE ICC Best Paper Award. She was the guest editor of *IEEE Journal on Selected Areas in Communications* (JSAC) Special Issue on Semantic Communications and area editor of IEEE JSAC Series. She was also the symposium co-chair of various flagship conferences. She was an associate editor for *IEEE Transactions on Communications*, *IEEE Transactions on Cognitive Networking*, and *IEEE Communications Letters*. She is also an area editor of *IEEE Communications Letters* and associate editor for *IEEE Signal Processing Magazine*.



Yanru Zhang (Senior Member, IEEE) received the BS degree in electronic engineering from the University of Electronic Science and Technology of China (UESTC) in 2012, and the PhD degree from the Department of Electrical and Computer Engineering, University of Houston (UH) in 2016. She was a postdoctoral fellow with UH and Chinese University of Hong Kong. She is currently a professor with the Shenzhen Institute for Advanced Study and School of Computer Science, UESTC. Her current research interests include game theory, machine learning, and deep learning in network economics, Internet, smart grid and applications, wireless communications, and networking. She was the recipient of Best Paper Award at IEEE HPCC 2022, DependSys 2022, ICC 2017, and ICCS 2016.



Zhu Han (Fellow, IEEE) received the BS degree in electronic engineering from Tsinghua University, in 1997, and the MS and PhD degrees in electrical and computer engineering from the University of Maryland, College Park, in 1999 and 2003, respectively. From 2000 to 2002, he was an R&D engineer with JDSU, Germantown, Maryland. From 2003 to 2006, he was a research associate with the University of Maryland. From 2006 to 2008, he was an assistant professor with Boise State University, Idaho. He is currently a John and Rebecca Moores professor with the Electrical and Computer Engineering Department and also with the Computer Science Department, University of Houston, Texas. His research interests include the novel game-theory related concepts critical to enabling efficient and distributive use of wireless networks with limited resources, wireless resource allocation and management, wireless communications and networking, quantum computing, data science, smart grid, carbon neutralization, security and privacy. Dr. Han was the recipient of NSF Career Award in 2010, Fred W. Ellersick Prize of the IEEE Communication Society in 2011, EURASIP Best Paper Award for the Journal on Advances in Signal Processing in 2015, IEEE Leonard G. Abraham Prize in the field of Communications Systems (best paper award in IEEE JSAC) in 2016, IEEE Vehicular Technology Society 2022 Best Land Transportation Paper Award, and several best paper awards in IEEE conferences. He was an IEEE Communications Society distinguished lecturer from 2015 to 2018 and ACM distinguished speaker from 2022 to 2025, AAAS fellow since 2019, and ACM Fellow since 2024. He has been a 1% highly cited researcher since 2017 according to Web of Science. He is also the winner of 2021 IEEE Kiyo Tomiyasu Award (an IEEE Field Award), for outstanding early to mid-career contributions to technologies holding the promise of innovative applications, with the following citation: "for contributions to game theory and distributed management of autonomous communication networks."